Warranty Revenue Forecasting System

Multi-Spectrum Forecasting & Decision Support

Yang Hsin En (William)
Data Analyst / Forecasting System Designer
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Problem

Warranty revenue decisions are often reactive.
Insights are based on historical reports only.

This leads to:

- Delayed response to revenue fluctuations
- Inefficient inventory and resource allocation
- Lack of forward-looking visibility

Objective

<u>Shift from reporting → forecasting → **decision support**</u>

Build a system that:

- Predicts future revenue
- Quantifies uncertainty
- Supports proactive decisions

01

Solution

A Multi-Spectrum Forecasting System that:

1. Combines multiple models

2. Captures different data patterns

3. Produces robust and reliable forecasts

A system that doesn't rely on one model.

But integrates multiple perspectives
to understand how the system behaves

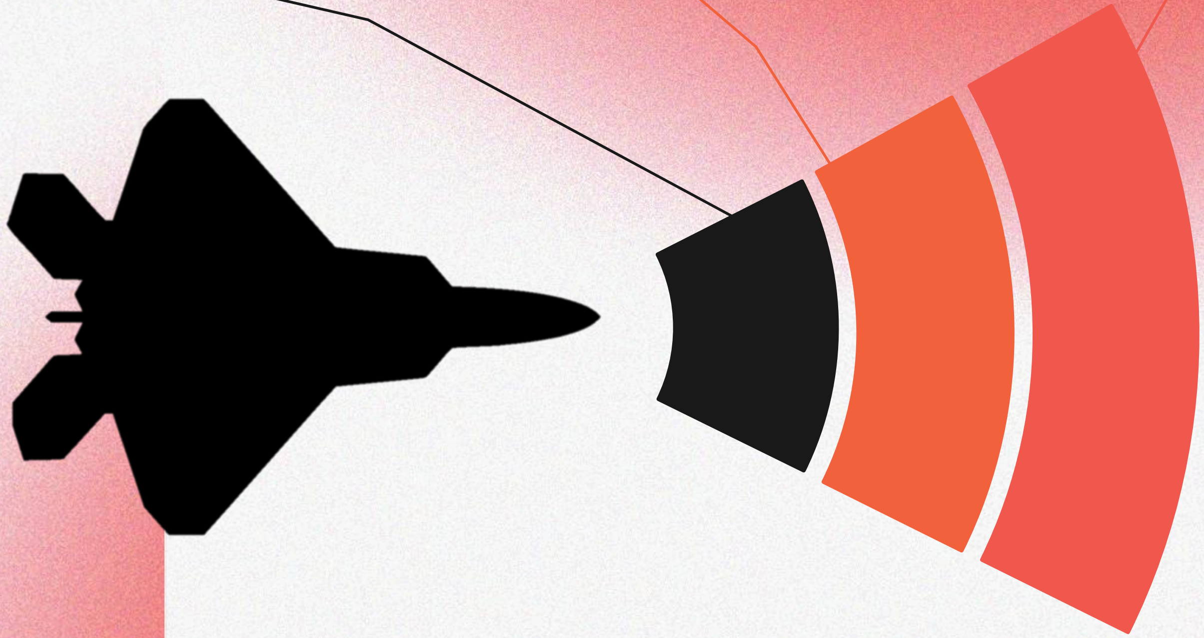
02 Architecture

Data induction

ETL process

Data Source
- Historical warranty claims data over past 3 years(29,000 obs.)
- Transaction-level records
- Aggregated into monthly revenue by branch

03 Multi-Spectrum Forecasting Framework



04 Feature Engineering

Instead of raw data,
the model learns structured signals from time series behavior

Trend
Direction

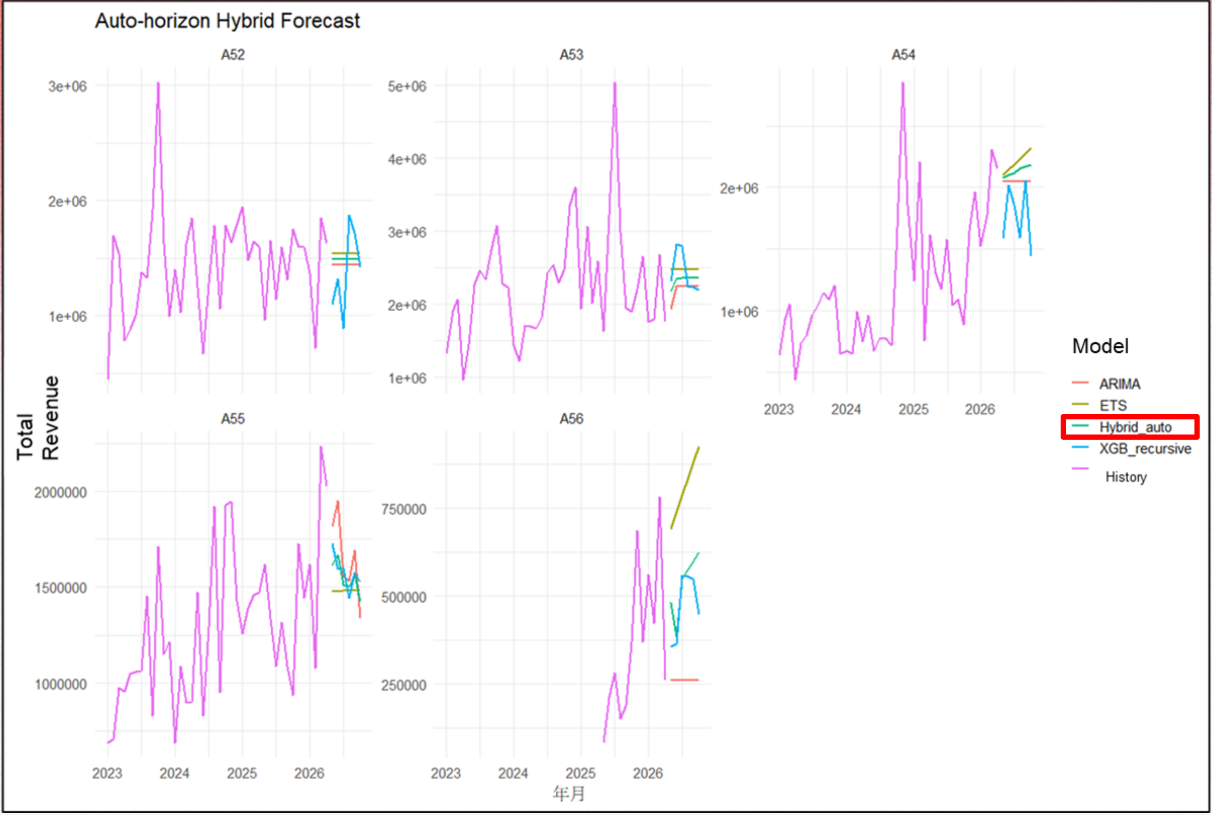
Growth
Momentum

Volatility
Stability

Seasonality
Cyclic Pattern

05 Result

Provides forecast with uncertainty range
Identifies high-risk periods
Enables risk-aware decision making



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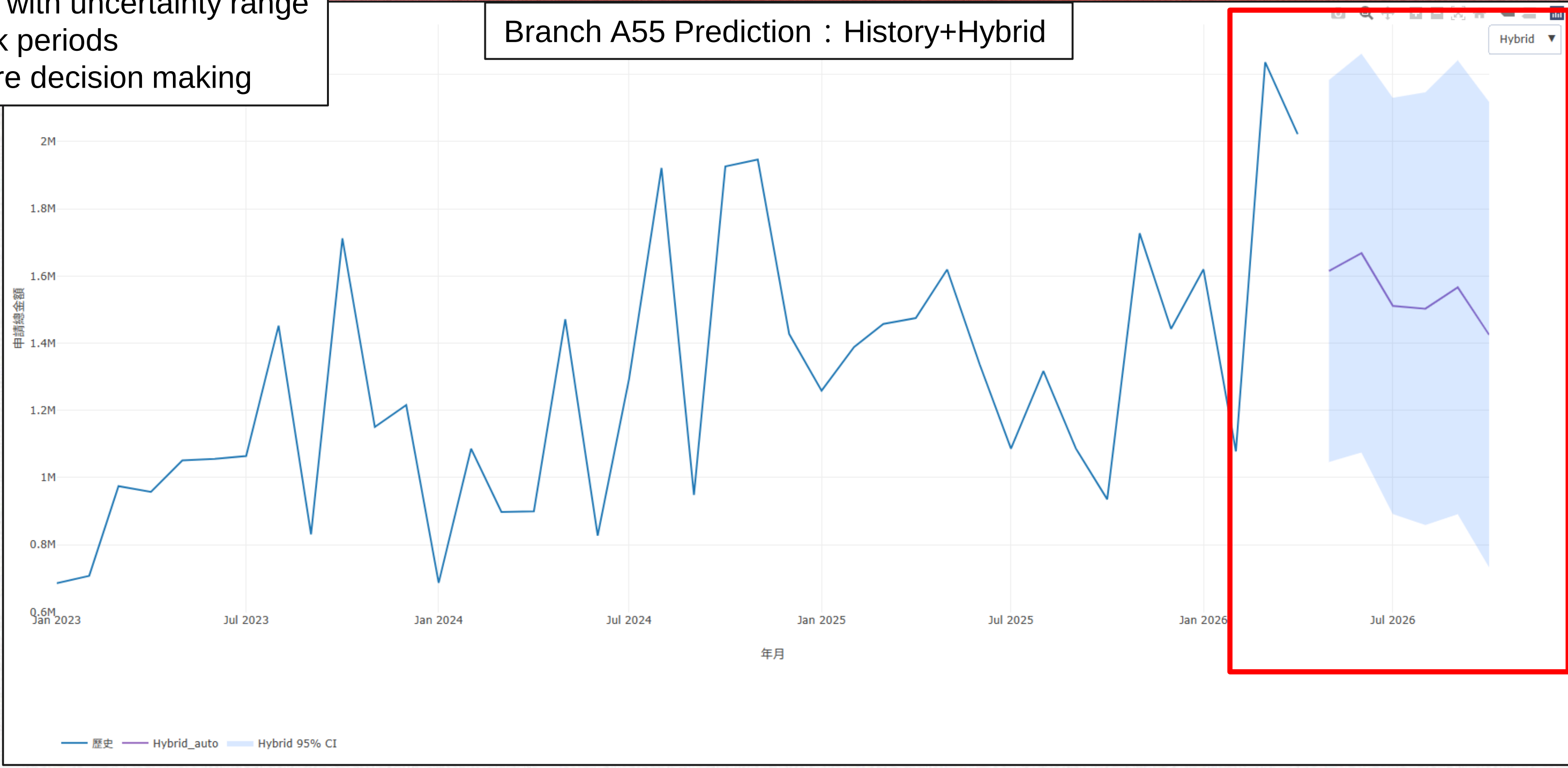
Branch A55 Prediction：History+Hybrid



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Decision Impact

The system enables:

- Proactive revenue planning
- Risk detection through uncertainty
- Better inventory and resource decisions

Key insight:
High uncertainty ≠ stable situation

Thank You

Contact：

willliam4785@gmail.com

GitHub：

https://github.com/william4785-code/Warranty-Revenue-Forecasting-System--Multi-Spectrum-Forecasting-Decision-Support-Framework-.git